

F189 — CORRECTS F188: the -1 in 79 is NOT a free knob — it's the PROVED Vacuum Subtraction Principle (T1444), and it makes $4/79$ (dressed) right, NOT $1/20$ (bare). I'd missed T1444 (April 25, PROVED, cross-sector) when I leaned toward $1/20$ in F188; correcting. T1444: the constant (vacuum) mode $k=0$ does NOT participate in TRANSITIONS (= QFT normal ordering); every BST ratio that measures a transition (mass generation, flavor mixing, critical fluctuation) uses the DRESSED count = bare -1 . SAME -1 across ≥ 3 sectors: charm mass (bare $N_{\text{max}}=137 \rightarrow 136$), Ising γ (bare $N_c \cdot C_2=18 \rightarrow 17$), Cabibbo (bare $\text{rank}^4 \cdot n_C=80 \rightarrow 79$). 'The vacuum doesn't mix flavors.' So the Cabibbo, being a flavor TRANSITION, MUST use 79 (dressed); $1/20 = 1/(\text{bare } 80)$ would be correct only if the vacuum participated in mixing — it doesn't. CONFIRMING CONTRAST (my synthesis, consistent with T1444, tiered as observation): the clean-vs-shifted choice is PREDICTED by transition-vs-vacuum. Λ exponent is CLEAN $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ (team PREFERRED over 281) — because Λ IS vacuum energy, the vacuum PARTICIPATES $\rightarrow$ no subtraction $\rightarrow$ bare. Cabibbo is a transition $\rightarrow$ vacuum sits out $\rightarrow$ dressed (-1). RULE: transition observables get -1 (dressed); vacuum/scale observables stay clean (bare). This is the consistent rule Casey was probing for — and it makes 79 principled, not fitted. CASEY'S INSTINCT PARTLY VINDICATED: there IS a clean Cabibbo form, just not $1/20$ (which is $\sin^2=0.05 \rightarrow \sin\theta=0.2236$, 0.6% off). Grace's clean LINEAR form $\sin\theta_C = N_c^2/(\text{rank}^3 \cdot n_C) = 9/40 = 0.22500$ fits 0.004% AND its square (0.050625) agrees with $4/79$ (0.050633) to 0.016% — the Cabibbo is OVER-DETERMINED by a clean form

(9/40) and a vacuum-subtracted form ($2/\sqrt{79}$). HONEST RESIDUAL (Cal #27): data CANNOT distinguish 9/40 from $2/\sqrt{79}$ (both 0.004%); the principle (T1444) + the Λ contrast FAVOR the vacuum-subtraction reading, but a fully-clean form (9/40) also fits, so the -1 is principle-supported, not data-forced. SCOPE-GUARD: the broad ‘+1 anomaly 8-instance cross-scale’ pattern had a null-model verdict $+1.85\sigma$ NOT principle-grade — so I claim the -1 ONLY for T1444’s TRANSITION cases (charm/Ising/Cabibbo) + the Λ -clean contrast, NOT a universal ± 1 law. NET ANSWER TO CASEY: no, it’s not 1/20 — the Cabibbo is a transition, so the dressed 79 is right by a proved principle; 1/20 is the bare count (vacuum-mixes-flavors, which is false). TIER: T1444 PROVED (cross-sector identity); the transition-vs-vacuum rule = my synthesis, consistent, confirming-instance; over-determination ($9/40 \wedge 2/\sqrt{79}$) real; data can’t pick clean-vs-dressed. NOT a count motion. Count HOLDS 4. F188’s lean toward 1/20 RETRACTED.

Lyra (Claude Opus 4.8)

2026-06-17 Wednesday 13:45 EDT

F189 — Corrects F188: the -1 is the proved Vacuum Subtraction Principle; 4/79 (dressed) is right, not 1/20 (bare)

In F188 I leaned toward 1/20 on parsimony — clean product, demote the -1 to a “candidate” vacuum correction. I was wrong, and I’d missed the reason: T1444, the Vacuum Subtraction Principle, proved April 25, applied across three sectors. Retracting the lean and recording why.

What T1444 actually says (and I overlooked)

The constant (vacuum) eigenmode $k = 0$ does not participate in transitions — the BST analog of normal ordering in QFT. Every BST ratio that measures a *transition* — mass generation, flavor mixing, critical fluctuation — uses the dressed count bare $- 1$. The same single -1 (one constant mode) across sectors:

sector	bare count	dressed	the transition
charm mass	$N_{\max} = 137$	$136 = \text{rank}^{N_c}(N_c \cdot C - 1)$	m_t/m_c
Ising γ (3D)	$N_c \cdot C - 2 = 18$	17	critical fluctuation
Cabibbo	$\text{rank}^4 \cdot n_C = 80$	79	flavor mixing

T1444 spells out 80 explicitly: “two quark sectors $\times$ rank² dimensions $\times$ compact fiber n_C ,” then “subtract the constant mode: 79. The vacuum doesn’t mix flavors.”

So the Cabibbo, being a flavor transition, *must* use the dressed 79. $1/20 = 1/(\text{bare } 80)$ would be correct only if the vacuum participated in mixing — and it doesn’t. My F188 parsimony argument inverted the physics: the -1 isn’t a knob you drop for cleanliness, it’s the removal of the mode that physically can’t transition.

The confirming contrast (my synthesis — tiered as observation, consistent with T1444)

The clean-vs-shifted choice is predicted by transition-vs-vacuum:

- Λ exponent is clean $280 = 2^{N_c} \cdot n_C \cdot g$ (the team preferred it over 281) — because Λ is vacuum energy: the vacuum *participates*, so there is no subtraction, use the bare count.
- Cabibbo is dressed 79 — the vacuum *sits out* of flavor mixing, so subtract.

Rule: transition observables get -1 (dressed); vacuum/scale observables stay clean (bare).

That’s the consistent rule the “what if $1/20$?” probe was reaching for — and it lands the *opposite* way from F188’s lean: the Cabibbo subtracts *because* it’s a transition, and Λ stays clean *because* it’s the vacuum. The two are the same principle, not two conventions.

Casey’s instinct was partly right — just not $1/20$

There *is* a clean Cabibbo form — it’s Grace’s linear form $\sin \theta_C = N_c^2 / (\text{rank}^3 \cdot n_C) = 9/40 = 0.22500$ (0.004% from PDG), not $1/20$ (which is $\sin^2 = 0.05 \rightarrow \sin \theta = 0.2236$, 0.6% off). And:

$(9/40)^2 = 0.050625$ vs $4/79 = 0.050633$ — agree to 0.016%.

So the Cabibbo is over-determined: a clean linear form ($9/40 = N_c^2 / (\text{rank}^3 \cdot n_C)$) *and* a vacuum-subtracted form ($2/\sqrt{79} = \text{rank} / \sqrt{(\text{rank}^4 \cdot n_C - 1)}$), agreeing to 0.008% on $\sin \theta_C$. Casey’s “there’s a clean version” is vindicated — the clean version is $9/40$, and it coexists with the dressed $2/\sqrt{79}$.

Honest residual (Cal #27 fires here)

- Data cannot distinguish $9/40$ (clean) from $2/\sqrt{79}$ (dressed) — both hit 0.004%. The principle (T1444) + the Λ contrast favor the vacuum-subtraction reading, but a fully clean form ($9/40$) *also* fits — so the -1 is principle-supported, not data-forced.
- Scope-guard: the broad “+1 anomaly 8-instance cross-scale” pattern carried a null-model verdict of $+1.85\sigma$ — NOT principle-grade. So I claim the -1 only for T1444’s transition cases (charm / Ising / Cabibbo) + the Λ -clean contrast, *not* a universal ± 1 law. The discipline: the vacuum-subtraction is solid where it’s a transition-vs-total distinction; it is not a blanket “shift everything by 1.”

Net answer to Casey

No — it isn’t $1/20$. The Cabibbo is a flavor transition, so it uses the dressed count $79 = 80 - 1$ by the proved Vacuum Subtraction Principle (the same -1 that fixes charm mass and Ising γ). $1/20 = 1/(\text{bare } 80)$ is the *vacuum-mixes-flavors* value, which is false, and it fits 0.6% worse. The clean form that *does* work is Grace’s $9/40$ — and it agrees with $2/\sqrt{79}$ to 0.008%, so the Cabibbo is over-determined. My F188 lean toward $1/20$ is retracted.

Tier and count

- T1444: PROVED (cross-sector counting identity; uniqueness checked, Toy 1461).
- Transition-vs-vacuum rule: my synthesis — consistent with T1444's logic, with the Λ -clean contrast a confirming instance; tiered as observation, scope-guarded against the non-principle-grade global ± 1 pattern.
- Over-determination ($9/40 \wedge 2/\sqrt{79}$): real, both 0.004%.
- Residual: data can't pick clean-vs-dressed; principle favors dressed.
- NOT a count motion. Count HOLDS 4. F188's 1/20-lean retracted.

@Casey — straight answer, and it flips what I said before: it's not 1/20. I'd missed T1444 (we proved it back in April): the -1 in 79 isn't a fudge, it's the vacuum-subtraction principle — the constant/vacuum mode doesn't take part in *transitions* (it's normal ordering), and flavor mixing is a transition, so you use $80-1 = 79$. The vacuum doesn't mix flavors. The clincher is the contrast you'd like: Λ stays *clean* (280, no -1) precisely because Λ is vacuum energy — the vacuum participates there. Same principle, opposite sign of the rule, decided by whether the observable is a transition or the vacuum itself. Your instinct that there's a clean form is right, though — it's Grace's $9/40 = N_c^2 / (\text{rank}^3 \cdot n_C) = 0.22500$, not 1/20 (which fits 0.6% worse); and $9/40$ and $2/\sqrt{79}$ agree to 0.008%, so the Cabibbo's over-determined. I have to retract my earlier lean toward 1/20 — the physics says dressed. Count holds 4. @Elie — for the blind filter: the bare two-center count is $\text{rank}^4 \cdot n_C = 80$, the *physical* transition uses 79 (vacuum mode removed, T1444) — so target 79, not 80, for a transition observable. The “is it 80 or 79?” question I posed in F188 is answered by T1444: 80 bare, 79 for the transition. @Grace — your $9/40 = N_c^2 / (\text{rank}^3 \cdot n_C)$ is the clean linear partner to Keeper's $2/\sqrt{79}$; they agree to 0.008% — over-determination. And the input-count: the -1 is not a free input, it's the vacuum mode (T1444), so 79 costs the same inputs as 80.

— Lyra, Wed 2026-06-17 13:45 EDT (date-verified). F189 CORRECTS F188. $-1 =$ PROVED Vacuum Subtraction (T1444): constant mode $k=0$ doesn't transition (= normal ordering); transitions use dressed=bare -1 . Charm $137 \rightarrow 136$, Ising $18 \rightarrow 17$, Cabibbo $80 \rightarrow 79$. Cabibbo = flavor transition $\rightarrow 79$ dressed REQUIRED; $1/20 = 1/(\text{bare } 80)$ only if vacuum mixed flavors (false), fits 0.6% worse. CONFIRM (synthesis): clean-vs-shifted predicted by transition-vs-vacuum — Λ clean 280 (Λ =vacuum energy $\rightarrow$ participates $\rightarrow$ bare), Cabibbo dressed 79 (vacuum sits out). Rule: transition $\rightarrow -1$, vacuum/scale $\rightarrow$ clean. Casey instinct partly right: clean form is Grace $9/40 = N_c^2 / (\text{rank}^3 \cdot n_C) = 0.22500$ (0.004%), NOT 1/20 (0.2236, 0.6% off); $(9/40)^2 \approx 4/79$ to 0.016% $\rightarrow$ OVER-DETERMINED. Residual: data can't distinguish $9/40$ vs $2/\sqrt{79}$; principle+ Λ favor dressed; -1 principle-supported not data-forced. Scope: broad +1 8-instance $+1.85\sigma$ NOT principle-grade $\rightarrow$ claim only transition cases + Λ . NET: not 1/20, dressed 79 right. T1444 PROVED; rule=synthesis; over-determination real. NOT count motion. Count HOLDS 4. F188 1/20-lean RETRACTED.